

Matrix Operations in Neural Networks:

- Addition & Subtraction
 - Transpose
 - Scalar Multiplication
 - Element-wise Multiplication
 - **Matrix Multiplication**
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1. Addition & Subtraction:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}_{(2 \times 2)} \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 5 \end{bmatrix}_{(2 \times 2)}$$

- shapes must match
- element-wise addition

$$A+B = \begin{bmatrix} 2+1 & 3+0 \\ 4+2 & 1+5 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 6 & 6 \end{bmatrix}_{(2 \times 2)}$$

2. Transpose:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{(2 \times 3)} \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{(3 \times 2)}$$

Flip rows $\leftrightarrow$ columns

3. Scalar Multiplication:

$$\alpha = 0.01 ; W = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- Scales every entry in a matrix

$$\alpha \cdot W = \begin{bmatrix} 0.01 \times 1 & 0.01 \times 2 \\ 0.01 \times 3 & 0.01 \times 4 \end{bmatrix} = \begin{bmatrix} 0.01 & 0.02 \\ 0.03 & 0.04 \end{bmatrix}$$

4. Element-wise multiplication:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 5 \end{bmatrix}$$

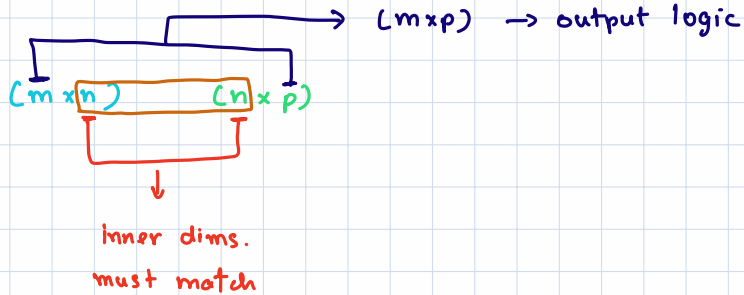
$$A \odot B = \begin{bmatrix} 2 \times 1 & 3 \times 0 \\ 4 \times 2 & 1 \times 5 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 8 & 5 \end{bmatrix}$$

5. Matrix Multiplication:

- Main computation inside neural networks ($W \cdot x$)

$A (m \times n)$

$B (n \times p)$



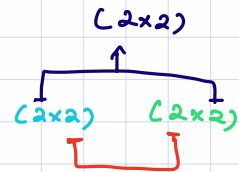
rule:

- inner dims must match
- outer dims gives output shape
- multiplication logic : row of A $\times$ column of B

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{(2 \times 2)}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{(2 \times 2)}$$



$$A \times B = \begin{bmatrix} (1 \ 2) \begin{bmatrix} 5 \\ 7 \end{bmatrix} & (1 \ 2) \begin{bmatrix} 6 \\ 8 \end{bmatrix} \\ (3 \ 4) \begin{bmatrix} 5 \\ 7 \end{bmatrix} & (3 \ 4) \begin{bmatrix} 6 \\ 8 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} (1 \times 5) + (2 \times 7) & (1 \times 6) + (2 \times 8) \\ (3 \times 5) + (4 \times 7) & (3 \times 6) + (4 \times 8) \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}_{(2 \times 2)}$$

Usage in DL:

- (*) $w \cdot x$ → forward pass → Mat. mul
 - (*) $w \cdot x + b$ → adding bias → Mat. add
 - (*) $\alpha \cdot \nabla w$ → apply learning rate → Scalar mult.
 - (*) $a \odot b$ → dropout → element-wise mult.
 - (*) w^T → backpropagation → Transpose
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